Advanced Bond Risk Research

Illustrative-book research unless a verified instrument master is supplied. Treasury CMT yields are zero-rate proxies; IG/HY index OAS are additive spread proxies. No market-price calibration, business-day adjustment, default, MBS optionality, FX or regulatory compliance.
Liquidity capacities and bid-ask inputs are assumptions.
Cash-short hedges are continuous, not integer futures trades.
PCA is full-sample descriptive analysis only.
ES diagnostics are not a formal ES backtest.
No production or regulatory approval is implied.

Research results require interpretation; model inventory entries are not approval or independent validation.

bond metrics.csv

id	dirty_price_per_100	clean_price_per_100	effective_rate_duration	effective_rate_convexity	rate_DV01
DEMO_UST2	98.2441044094587	98.2441044094587	1.943757846464925	3.8441149083324535	2864.44123222207
DEMO_UST5	95.01726549003482	95.01726549003482	4.568407304545721	22.084973433506025	8681.551394452703
DEMO_UST10	93.64966351807662	93.64966351807662	8.07949705491967	74.62206665786813	15132.843611770364
DEMO_UST30	83.77329183932684	83.77329183932684	15.292702794456622	345.33149249323947	19216.800813181555
DEMO_IG7	96.65306216828316	96.65306216828316	5.878985975979147	38.5328223253958	11364.439940455552
DEMO_HY5	96.55769116857468	96.55769116857468	4.244024935276841	19.91508628381396	4097.932490121914

concentration.csv

id	issuer	sector	credit_bucket	market_value	weight
DEMO_UST2	ILLUSTRATIVE	Government	GOV	14736615.661418805	0.1567351960737893
DEMO_UST5	ILLUSTRATIVE	Government	GOV	19003453.098006964	0.202116280686689
DEMO_UST10	ILLUSTRATIVE	Government	GOV	18729932.703615323	0.199207181770756
DEMO_UST30	ILLUSTRATIVE	Government	GOV	12565993.775899028	0.133648969585594
DEMO_IG7	ILLUSTRATIVE	Corporate	IG	19330612.43365664	0.2055958708313019
DEMO_HY5	ILLUSTRATIVE	Corporate	HY	9655769.11685747	0.1026965010518696

risk validation.csv

observations	exceptions	exception_rate	Kupiec_LR	Kupiec_p	Christoffersen_ind_LR
531	5	0.0094161958568738	0.0186435432371823	0.8913933294328736	0.0952395350197008
531	5	0.0094161958568738	0.0186435432371823	0.8913933294328736	0.0952395350197008
531	5	0.0094161958568738	0.0186435432371823	0.8913933294328736	0.0952395350197008
531	10	0.0188323917137476	3.321832487060547	0.0683655870485916	0.3846390946924884

observations	Christoffersen_ind_p	conditional_coverage_p	model	mean_quantile_loss	mean_loss_on_exceptions
531	0.7576189490083254	0.944649292773276	Gaussian	0.0001042166127111	0.0091474102875724
531	0.7576189490083254	0.944649292773276	Historical	0.0001058460051442	0.0091474102875724
531	0.7576189490083254	0.944649292773276	Student_t_df5	0.0001065118894658	0.0091474102875724
531	0.5351308688741139	0.1567292020017767	Vol_scaled_historical	0.0001087063314859	0.0087299820166703

observations	mean_ES_on_exceptions
531	0.008070914863736
531	0.0086570155162532
531	0.0104826460725938
531	0.008389938132013

mc summary.csv

horizon	confidence	VaR	ES	paths	excluded
1	0.95	0.0056887385939787	0.0072303271316297	1000	Carry, time roll, defaults and transaction costs
1	0.975	0.006907286263843	0.008182492954635	1000	Carry, time roll, defaults and transaction costs
1	0.99	0.0083223263040377	0.0093017893109588	1000	Carry, time roll, defaults and transaction costs
10	0.95	0.0170809036570936	0.0216558343635461	1000	Carry, time roll, defaults and transaction costs
10	0.975	0.0203351110179101	0.0246847178357253	1000	Carry, time roll, defaults and transaction costs
10	0.99	0.0245637152680135	0.0285514353720794	1000	Carry, time roll, defaults and transaction costs

hedge trades.csv

id	short_face_usd	implementation
DEMO_UST2	16309598.418978872	Continuous cash-bond short; not futures
DEMO_UST5	30897940.41356725	Continuous cash-bond short; not futures
DEMO_UST10	20027208.360279854	Continuous cash-bond short; not futures
DEMO_UST30	15000000.0	Continuous cash-bond short; not futures

hedge risk.csv

factor	before_usd_per_bp	after_usd_per_bp
DGS1MO	0.0	0.0
DGS3MO	1.90263784956187	0.5319043374714376
DGS6MO	113.84116463176906	31.82560954977663
DGS1	397.6595690874383	109.8064917357142
DGS2	3545.165560942143	0.0
DGS3	1741.091941356659	549.0674173114
DGS5	13733.669026958756	0.0
DGS7	11748.015297034755	9769.931448571817
DGS10	15403.851662364788	0.0
DGS20	4302.154284282588	-0.013479845079928
DGS30	10370.655099176802	0.0
IG	11364.4399404563	11364.4399404563
HY	4097.9324901225045	4097.9324901225045

stress results.csv

scenario	pnl_usd	pnl_pct	type
Rates_up_100bp	-5799318.426172614	-6.168019384533449	Hypothetical; no assigned probability
Rates_down_100bp	6517114.936169148	6.931450957427959	Hypothetical; no assigned probability
Steepener	-3527708.4137431234	-3.751988126871943	Hypothetical; no assigned probability
Flattener	732696.8913396448	0.7792792698491173	Hypothetical; no assigned probability
Credit_stress	-2246916.060069025	-2.389767347724658	Hypothetical; no assigned probability
Combined_stress	-7924180.864487976	-8.427973355994519	Hypothetical; no assigned probability

reverse stress result.csv

target_loss_usd	achieved_loss_usd	solver_success	constraint_satisfied	message	interpretation
9402237.678945422	9402236.207753718	True	True	Optimization terminated successfully	Local constrained solution; diagonal standardized distance is not a probability or global minimum guarantee

liquidity stress.csv

outflow_fraction	days	required_cash	net_liquidation_capacity	shortfall	assumption
0.1	1	9402237.678945422	41945000.0	0.0	User/demo capacities; constant prices; simultaneous liquidation; no impact feedback
0.1	5	9402237.678945422	74895959.24370119	0.0	User/demo capacities; constant prices; simultaneous liquidation; no impact feedback
0.1	10	9402237.678945422	84479310.09218223	0.0	User/demo capacities; constant prices; simultaneous liquidation; no impact feedback
0.25	1	23505594.197363555	41945000.0	0.0	User/demo capacities; constant prices; simultaneous liquidation; no impact feedback
0.25	5	23505594.197363555	74895959.24370119	0.0	User/demo capacities; constant prices; simultaneous liquidation; no impact feedback
0.25	10	23505594.197363555	84479310.09218223	0.0	User/demo capacities; constant prices; simultaneous liquidation; no impact feedback
0.5	1	47011188.39472711	41945000.0	5066188.394727111	User/demo capacities; constant prices; simultaneous liquidation; no impact feedback
0.5	5	47011188.39472711	74895959.24370119	0.0	User/demo capacities; constant prices; simultaneous liquidation; no impact feedback
0.5	10	47011188.39472711	84479310.09218223	0.0	User/demo capacities; constant prices; simultaneous liquidation; no impact feedback

interval evaluation.csv

period	nominal_coverage	observed_coverage	mean_width	raw_crossing_fraction	baseline_RMSE
validation	0.9	0.6666666666666666	0.0034564860218893	0.0	0.0008046896785515
test	0.9	0.9027777777777778	0.0037262259651397	0.0	0.001127155148185

period	ML_RMSE
validation	0.0014485171554599
test	0.0015503294716009

advanced policy metrics.csv

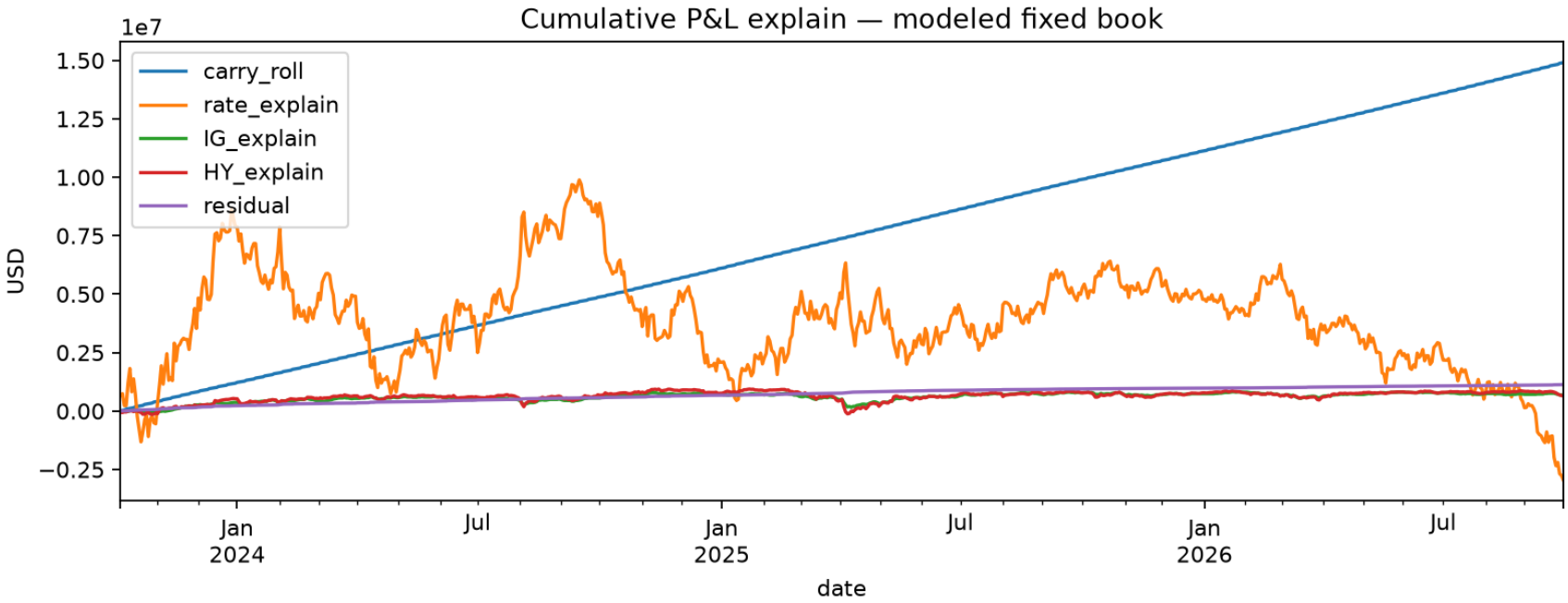
strategy	sum_normalized_hpl	daily_std	worst_day	total_cost	turnover_usd
MC_RL	0.0307957333055822	0.0018947646095334	-0.0043814289923408	0.0023291583062704	1175000000.0
No_hedge	0.0458992968752413	0.0023735659584483	-0.0059132655734225	0.0	0.0
Static_long	0.0318500852414287	0.0016474732414828	-0.0039799877649504	9.649922110143802e-05	25000000.0
Vol_rule	0.0458992968752413	0.0023735659584483	-0.0059132655734225	0.0	0.0
MC_RL	-0.0422879154045233	0.0026019418075286	-0.007568827862642	0.0053015423291844	2625000000.0
No_hedge	-0.0403538357149441	0.0030940170736438	-0.0093425076783541	0.0	0.0
Static_long	-0.0265304886866402	0.0021516803654367	-0.0064167784789712	9.948790928117496e-05	25000000.0
Vol_rule	-0.0384994692703673	0.0029383240974222	-0.0093932833814845	0.0015051765673434	725000000.0
MC_RL	0.0296024151036239	0.0018081334296267	-0.0040766680241363	0.0089085315856061	4575000000.0
No_hedge	0.0458992968752413	0.0023735659584483	-0.0059132655734225	0.0	0.0
Static_long	0.0318500852414287	0.0016474732414828	-0.0039799877649504	9.649922110143802e-05	25000000.0
Vol_rule	0.0458992968752413	0.0023735659584483	-0.0059132655734225	0.0	0.0
MC_RL	-0.0378172238102267	0.0023760244176048	-0.0076192479708111	0.0052848511384728	2650000000.0
No_hedge	-0.0403538357149441	0.0030940170736438	-0.0093425076783541	0.0	0.0
Static_long	-0.0265304886866402	0.0021516803654367	-0.0064167784789712	9.948790928117496e-05	25000000.0
Vol_rule	-0.0384994692703673	0.0029383240974222	-0.0093932833814845	0.0015051765673434	725000000.0
MC_RL	0.0267390965630478	0.0019235713302638	-0.0047321362840551	0.0087152535010504	4475000000.0
No_hedge	0.0458992968752413	0.0023735659584483	-0.0059132655734225	0.0	0.0
Static_long	0.0318500852414287	0.0016474732414828	-0.0039799877649504	9.649922110143802e-05	25000000.0
Vol_rule	0.0458992968752413	0.0023735659584483	-0.0059132655734225	0.0	0.0
MC_RL	-0.0371204190792925	0.0024954203814438	-0.0072074077181627	0.0057973069732248	2875000000.0
No_hedge	-0.0403538357149441	0.0030940170736438	-0.0093425076783541	0.0	0.0
Static_long	-0.0265304886866402	0.0021516803654367	-0.0064167784789712	9.948790928117496e-05	25000000.0
Vol_rule	-0.0384994692703673	0.0029383240974222	-0.0093932833814845	0.0015051765673434	725000000.0

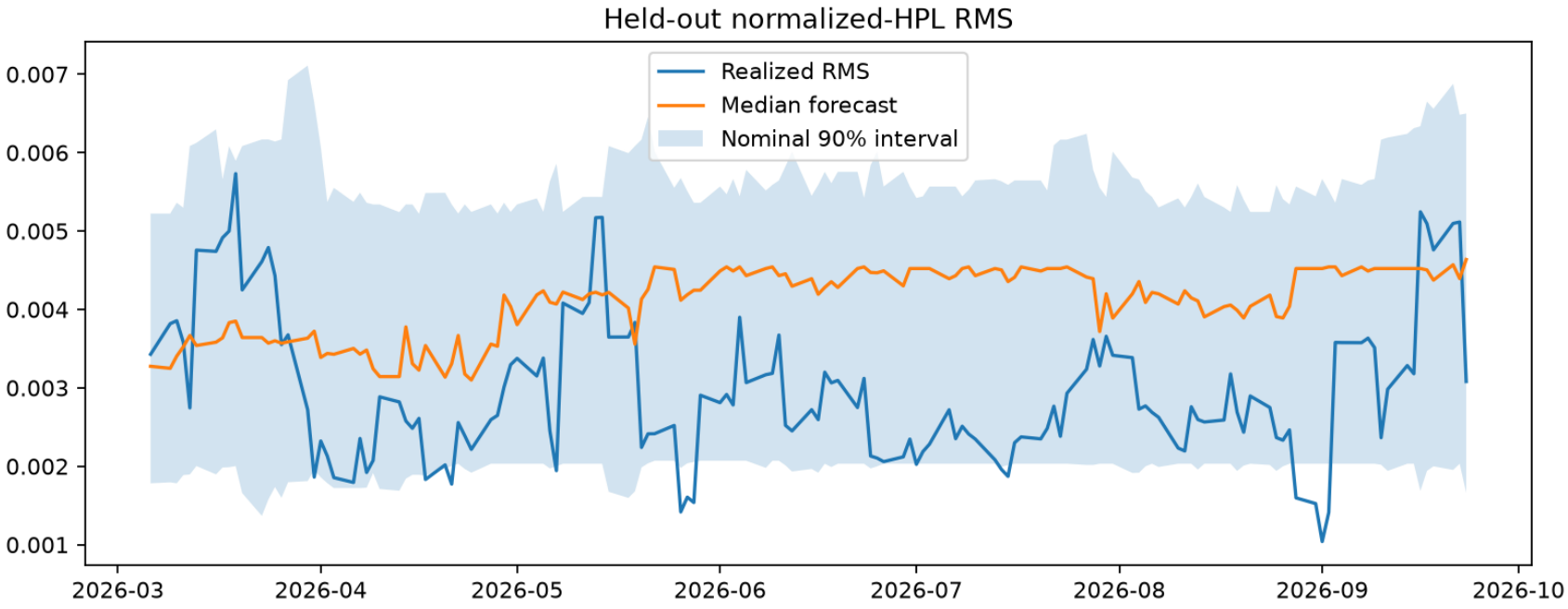
strategy	interpretation	period	seed
MC_RL	Normalized fixed-book HPL, not a funded NAV return	validation	7
No_hedge	Normalized fixed-book HPL, not a funded NAV return	validation	7
Static_long	Normalized fixed-book HPL, not a funded NAV return	validation	7
Vol_rule	Normalized fixed-book HPL, not a funded NAV return	validation	7
MC_RL	Normalized fixed-book HPL, not a funded NAV return	test	7
No_hedge	Normalized fixed-book HPL, not a funded NAV return	test	7

strategy	interpretation	period	seed
Static_long	Normalized fixed-book HPL, not a funded NAV return	test	7
Vol_rule	Normalized fixed-book HPL, not a funded NAV return	test	7
MC_RL	Normalized fixed-book HPL, not a funded NAV return	validation	42
No_hedge	Normalized fixed-book HPL, not a funded NAV return	validation	42
Static_long	Normalized fixed-book HPL, not a funded NAV return	validation	42
Vol_rule	Normalized fixed-book HPL, not a funded NAV return	validation	42
MC_RL	Normalized fixed-book HPL, not a funded NAV return	test	42
No_hedge	Normalized fixed-book HPL, not a funded NAV return	test	42
Static_long	Normalized fixed-book HPL, not a funded NAV return	test	42
Vol_rule	Normalized fixed-book HPL, not a funded NAV return	test	42
MC_RL	Normalized fixed-book HPL, not a funded NAV return	validation	123
No_hedge	Normalized fixed-book HPL, not a funded NAV return	validation	123
Static_long	Normalized fixed-book HPL, not a funded NAV return	validation	123
Vol_rule	Normalized fixed-book HPL, not a funded NAV return	validation	123
MC_RL	Normalized fixed-book HPL, not a funded NAV return	test	123
No_hedge	Normalized fixed-book HPL, not a funded NAV return	test	123
Static_long	Normalized fixed-book HPL, not a funded NAV return	test	123
Vol_rule	Normalized fixed-book HPL, not a funded NAV return	test	123

model inventory.csv

model	owner	status	intended_use	prohibited_use
Cashflow valuation	Project author	Research / not independently validated	Model-based fixed-income research	Live execution or regulatory capital
Rolling VaR	Project author	Research / not independently validated	Historical forecast comparison	Live execution or regulatory capital
Factor bootstrap	Project author	Research / not independently validated	Hypothetical market-shock risk	Live execution or regulatory capital
Cash hedge optimizer	Project author	Research / not independently validated	Sensitivity reduction study	Live execution or regulatory capital
Reverse stress	Project author	Research / not independently validated	Local loss-threshold scenario search	Live execution or regulatory capital
ML quantiles	Project author	Research / not independently validated	Volatility interval research	Live execution or regulatory capital
MC hedge RL	Project author	Research / not independently validated	Experimental policy evaluation	Live execution or regulatory capital


Cumulative P&L explain — modeled fixed book
1e7
1.50
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
USD
carry_roll
rate_explain
IG_explain
HY_explain
residual
Jan
2024
Jul
Jan
2025
Jul
Jan
2026
Jul
date


Held-out normalized-HPL RMS
Realized RMS
Median forecast
Nominal 90% interval
0.007
0.006
0.005
0.004
0.003
0.002
0.001
2026-03
2026-04
2026-05
2026-06
2026-07
2026-08
2026-09
2026-10